Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction

A Report Submitted
in Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology
in
Computer Science & Engineering

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November, 2026

UNDERTAKING

We declare that the work presented in this report titled "*Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be unconditionally withdrawn.

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CERTIFICATE

Certified that the work contained in the report titled "*Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction*", by Shaurya Pratap Singh (20233262), Surya Pratap Singh (20233283), Vishesh Vigh (20233583), and Sehar Majid Abdi (20233561), has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

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Abstract

Predicting whether and when a breast cancer will progress after treatment is a central question in oncology, and a prediction helps a clinician only if it comes with reasons. We build a knowledge graph for each of the 1,083 breast cancer patients of the TCGA PanCancer Atlas 2018 cohort. It links the patient to their tumour and to their altered genes, and we join those genes to PrimeKG, a public biomedical knowledge graph of pathways, drugs, and diseases.

Graph neural networks (GNNs) trained on these graphs reach a concordance index (C-index) of about 0.62, and every model family we tried stops near 0.67. We therefore turned the project towards explanation. A survival GNN supplies neural evidence. For the symbolic side we replaced crisp IF–THEN rules with learned logic. We compared four survival-aware reasoners: a Markov logic network whose formula weights are learned from survival times, probabilistic soft logic, rules written by a reinforcement-learning agent, and a hybrid in which reinforcement learning proposes the Markov logic formulas. The hybrid is our rule engine. Each explanation path is scored as $0.6\times$ neural + $0.4\times$ rule.

The logic engine matches RuleKit's accuracy (C-index 0.63 against 0.62). 69% of its weighted formulas reason over knowledge-graph relations, against 10% of RuleKit's rules. The fused model is statistically tied with the best clinical model (0.664 against 0.675). Its month-of-progression estimate is within a year for 48% of progressors. The system writes an explained PDF report for every patient, grouped by when they progressed.

Acknowledgements

We thank our supervisor, Dr. Shashwati Banerjea, for guidance and for steering the project from raw accuracy towards neuro-symbolic reasoning and interpretable patient reports. We thank the Department of Computer Science and Engineering, MNNIT Allahabad, for its support. We are grateful to the TCGA Research Network and cBioPortal for the data, to the Zitnik Lab for PrimeKG, and to the developers of PyTorch Geometric, lifelines, scikit-survival, and RuleKit.

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Chapter 1

Introduction and Background

Breast cancer is the most commonly diagnosed cancer worldwide [1]. After treatment, some patients never relapse, some progress within months, and some progress after a decade. Clinicians estimate this risk mainly from stage, lymph-node status, and molecular subtype [2]. Tumour sequencing adds hundreds of gene alterations per patient, but most of them mean little on their own.

A biomedical knowledge graph can put those alterations in context. PrimeKG [3] connects genes to the pathways they act in, the drugs that target them, and the diseases they are linked to. If a patient's altered gene sits in a growth-signalling pathway, the graph can say so explicitly.

1.1 Motivation

A risk score without reasons is hard to trust in a clinic. Neural networks learn well from data but do not explain themselves. Rules are readable but cannot weigh thousands of interacting facts. Neuro-symbolic AI [4] combines the two: a neural part learns from the data, and a symbolic part states its reasoning as explicit, checkable knowledge. Our project began as a GNN benchmark. When accuracy saturated, we followed our supervisor's direction and built a neuro-symbolic layer whose output is a readable report for every patient.

1.2 Problem Statement

Given a patient's clinical record, tumour mutations, and copy-number changes, together with PrimeKG, the system must (i) estimate the risk of progression, (ii) estimate the month of progression when it happens, (iii) explain the estimate with knowledge-graph paths and logic formulas, and (iv) present it as a clinical report grouped by progression time. Outcome data must never leak into the inputs. Every reported prediction must come from a model that did not see that patient during training.

1.3 Contributions and Novelty

1. A leakage-checked pipeline from raw TCGA tables to patient graphs fused with PrimeKG, with 58 unit tests.

2. A benchmark of seven GNN architectures and three training objectives, all evaluated out-of-fold.

3. **Learned logic instead of IF–THEN rules.** We tried all three of our supervisor's suggestions (Markov logic, probabilistic soft logic, and reinforcement learning) as survival reasoners over 422 first-order facts. The chosen engine is new in two ways:

    - it is a *survival* Markov logic network, whose formula weights are learned with the Cox likelihood, so patients who have not relapsed yet are handled correctly;
    - its formulas, many of them multi-step knowledge-graph chains, are *proposed by a reinforcement-learning agent* that is rewarded only for rules that hold up on unseen patients.

4. Explanation paths scored as $0.6\times$ neural importance $+ 0.4\times$ the weight of the formulas they support, with a top-K cutoff.

5. 1,083 patient reports in five progression groups, and two executed notebooks that document the work.

1.4 Background: Data and Knowledge Graph

The Cancer Genome Atlas (TCGA) PanCancer Atlas profiled more than 10,000 tumours with uniform pipelines [5]. We use its breast cancer cohort as distributed by cBioPortal [6]. Its progression-free interval is the recommended outcome for breast cancer [7]. PrimeKG [3] merges 20 biomedical resources, including Reactome pathways and DrugBank targets, into 129,375 nodes and over 8 million edges. Its genes keep their NCBI Entrez identifiers, so patient genes can be matched exactly.

1.5 Background: Graph Neural Networks

A GNN updates every node from its neighbours' features. Let $h_v^{(l)}$ be the vector of node v at layer l, $\mathcal{N}(v)$ its neighbours, and d_v its degree. GCN [8] averages neighbours with degree normalisation, and GraphSAGE [9] keeps separate self and neighbour information:

$$h_v^{(l+1)} = \sigma\Big(\sum_{u \in \mathcal{N}(v) \cup \{v\}} \frac{1}{\sqrt{d_u d_v}} \, W^{(l)} h_u^{(l)} \Big), \tag{1}$$

$$h_v^{(l+1)} = \sigma\Big(W^{(l)} \big[\, h_v^{(l)} \, \| \, \operatorname{mean}_{u \in \mathcal{N}(v)} h_u^{(l)} \big] \Big). \tag{2}$$

GATv2 [10] learns an attention weight for each edge, and R-GCN [11] learns one weight matrix per relation type r, with a normalising constant $c_{v,r}$:

$$\alpha_{vu} = \frac{\exp\big(a^\top \operatorname{LeakyReLU}(W[h_v \| h_u])\big)}{\sum_{k \in \mathcal{N}(v)} \exp\big(a^\top \operatorname{LeakyReLU}(W[h_v \| h_k])\big)}, \tag{3}$$

$$h_v^{(l+1)} = \sigma\Big(W_0^{(l)} h_v^{(l)} + \sum_{r} \sum_{u \in \mathcal{N}_r(v)} \frac{1}{c_{v,r}} W_r^{(l)} h_u^{(l)} \Big). \tag{4}$$

GNNExplainer [12] learns a soft edge mask M that keeps a trained model's prediction Y for graph G:

$$\min_{M} \; -\log P\big(Y \mid G \odot \sigma(M)\big). \tag{5}$$

The value $\sigma(M_e)$ is the importance of edge e.

1.6 Background: Survival Analysis

Progression is a time-to-event outcome with *censoring*: a patient with no recorded progression may still progress after their last visit. Each patient i has a time t_i and an event flag δ_i, which is 1 for a recorded progression and 0 for censoring.

The Kaplan–Meier curve [13] estimates the share of patients still progression-free, where d_i is the number of events at time t_i and n_i the number still at risk. The log-rank test [14] compares two groups through their observed events O_{1i}, expected events E_{1i}, and variance V_{1i}:

$$\hat{S}(t) = \prod_{t_i \le t} \Big(1 - \frac{d_i}{n_i}\Big), \qquad z = \frac{\sum_i (O_{1i} - E_{1i})}{\sqrt{\sum_i V_{1i}}}. \tag{6}$$

The Cox model [15] assumes a shared baseline hazard $h_0(t)$ scaled by a log-risk $\eta = \beta^\top x$. A coefficient β_j gives the hazard ratio $\exp(\beta_j)$. The model is fitted by the partial likelihood, which keeps censored patients in the risk sets for as long as they were followed:

$$h(t \mid x) = h_0(t)\, e^{\eta}, \qquad \mathcal{L}(\beta) = \prod_{i:\, \delta_i = 1} \frac{e^{\eta_i}}{\sum_{j:\, t_j \ge t_i} e^{\eta_j}}. \tag{7}$$

With the cumulative baseline hazard $H_0(t)$, a patient's survival curve is $S(t \mid x) = \exp\big(-H_0(t)\, e^{\eta}\big)$.

Accelerated failure time (AFT) models [16] predict the event time directly, $\log T = \mu(x) + \sigma\varepsilon$. With logistic noise, T is log-logistic with shape $\rho = 1/\sigma$, and a share p of patients remain event-free at

$$t_{(p)} = e^{\mu(x)} \Big(\frac{1}{p} - 1\Big)^{1/\rho}, \tag{8}$$

so the median is $e^{\mu(x)}$.

Models are compared with the concordance index (C-index) [17]. A pair is comparable when $\delta_i = 1$ and $t_i < t_j$, so we know that i progressed first. For risk scores r,

$$C = \frac{\sum_{(i,j)\ \text{comparable}} \big[\mathbf{1}(r_i > r_j) + \tfrac{1}{2}\mathbf{1}(r_i = r_j)\big]}{\#\{\text{comparable pairs}\}}. \tag{9}$$

A C-index of 0.5 is a coin flip and 1.0 is perfect. Uno's C [35] reweights pairs by the inverse probability of censoring $\hat{G}$. The integrated Brier score (IBS) averages the

censoring-weighted squared error of predicted survival up to a horizon τ, and lower is better:

$$\mathrm{IBS} = \frac{1}{\tau}\int_0^\tau \frac{1}{N}\sum_i \Big[\frac{\hat{S}(t \mid x_i)^2\,\mathbf{1}(t_i \le t, \delta_i = 1)}{\hat{G}(t_i)} + \frac{(1-\hat{S}(t \mid x_i))^2\,\mathbf{1}(t_i > t)}{\hat{G}(t)}\Big]dt. \quad (10)$$

1.7 Background: Rules and Logic

Survival rules. RuleKit's LR-Rules [18, 19] grows IF–THEN rules such as "IF nodal stage is N0 AND age $\geq$ 41.5 THEN low risk" by maximising the log-rank separation of the patients a rule covers. The rules are crisp: a rule either fires or it does not, and it carries no weight.

Markov logic networks (MLNs) [20] attach a weight w_k to each logic formula and define a probability over possible worlds x, where $n_k(x)$ counts the true groundings of formula k:

$$P(X = x) = \frac{1}{Z}\exp\Big(\sum_k w_k\, n_k(x)\Big). \quad (11)$$

A world that breaks a formula with a large weight becomes less likely but is still possible. Learning an MLN means finding the formulas (structure learning [36]) and their weights.

Probabilistic soft logic (PSL) [21] allows truth values between 0 and 1, with the Łukasiewicz operators

$$a \mathbin{\tilde{\wedge}} b = \max(a + b - 1, 0), \qquad a \mathbin{\tilde{\vee}} b = \min(a + b, 1), \qquad \tilde{\neg} a = 1 - a. \quad (12)$$

A rule $w : B \Rightarrow H$ becomes a penalty for how far it is from being satisfied, $\phi = w \max(0,\, B - H)^q$. Inference minimises the sum of these penalties, which is a convex optimisation. That allows *collective* reasoning, for example letting similar patients influence each other.

Reinforcement learning (RL) for rules. A policy-gradient agent learns from rewards rather than labels. With REINFORCE [22], a policy π_θ that samples a rule $a_{1:T}$ with reward R is updated by

$$\nabla_\theta J = \mathbb{E}\Big[(R - b)\sum_t \nabla_\theta \log \pi_\theta(a_t \mid a_{1:t-1})\Big] + \beta_H \nabla_\theta \mathcal{H}(\pi_\theta), \quad (13)$$

where b is a moving-average baseline and the entropy $\mathcal{H}$ keeps the agent exploring. In knowledge-graph reasoning, agents have been trained to walk relation paths [23, 24] and to write logic rules [25]. We use RL to write survival rules.

Chapter 2

Proposed Work

Figure 1 gives the overview. Patient graphs are built from TCGA and fused with PrimeKG. A survival GNN and the logic reasoners run on the same patients and the same five folds. Their risks are combined in one Cox model, and every patient gets a report that explains the prediction with knowledge-graph paths and logic formulas.

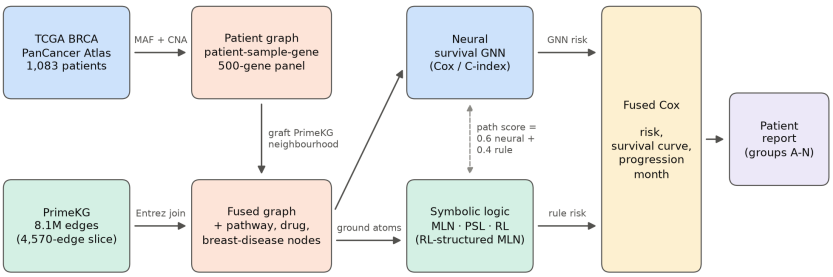


Figure 1: Overview of the neuro-symbolic progression system.

2.1 Data and Labels

We use the cBioPortal `brca_tcga_pan_can_atlas_2018` study, checked against its official checksums: the clinical tables, 126,252 mutation records, and copy-number calls for 25,128 genes. 1,083 patients have a usable progression-free survival (PFS) record, and 145 of them progressed. Outcome columns (survival, progression, and

tumour-status fields) are blocked from every feature table by a guard function that the tests check.

The GNN benchmark uses a 36-month landmark label (498 patients, 89 events). The survival models and the reports use the full follow-up time and event of all 1,083 patients.

2.2 Patient Graphs and PrimeKG Fusion

Each patient graph has the shape Patient → Sample → Gene. Genes are limited to a 500-gene panel of the most frequently altered genes plus 27 known breast cancer drivers. PrimeKG's 8.1 million edges are filtered once to 4,570 edges that touch the panel: protein–protein, gene–pathway, gene–drug, and gene–breast-disease links. A patient's genes join PrimeKG on their Entrez identifiers, which adds each altered gene's partners, pathways, drugs, and breast-disease links (Figure 2). The median fused graph has 632 nodes.

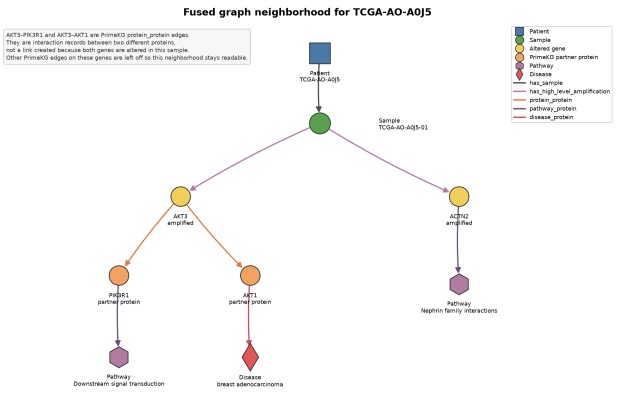


Figure 2: Part of a fused graph: a patient's altered genes linked to PrimeKG proteins, pathways, and a breast-disease node.

2.3 Neural Part: Graph Neural Networks

All GNNs share three message-passing layers of width 64, dropout 0.3, and a readout over the patient and altered-gene nodes. A linear head on the clinical block c_p is

added to the graph score through a gate g that starts near 0.12, so training begins from the clinical score:

$$r_p = w^\top c_p + g\, f_{\text{graph}}(G_p). \tag{14}$$

Three objectives were tested. On the landmark endpoint, models use weighted cross-entropy or a smooth C-index over the comparable pairs P of a mini-batch, with temperature τ_c. On the full follow-up, they use the Cox partial likelihood, as in DeepSurv [26]:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_i \big[w_+ y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\big], \tag{15}$$

$$\mathcal{L}_C = 1 - \frac{1}{|P|}\sum_{(i,j)\in P} \sigma\Big(\frac{r_i - r_j}{\tau_c}\Big), \tag{16}$$

$$\mathcal{L}_{\text{Cox}} = -\frac{1}{N_e}\sum_{i:\,\delta_i=1}\Big(r_i - \log\sum_{j:\,t_j\geq t_i} e^{r_j}\Big). \tag{17}$$

The best survival GNN supplies a neural risk. GNNExplainer, run on the fold model that held the patient out, gives every edge of the patient's graph an importance score.

2.4 Graph Structure Learning: Evaluated, Not Adopted

Graph structure learning (GSL) [33] learns which edges a GNN should use. Our version retrieved up to three hops and gave each PrimeKG edge e a learned weight, computed from its endpoints, its relation, and the patient's clinical context, that multiplies the message on that edge. The first version also let the scorer read the hop number. The second removed that input. Edges within two hops start at weight 1, and the third hop stays closed behind a patient-specific gate g_p that starts near 0.12 and must pass 0.5:

$$w_e = 2\,\sigma\big(\text{logit}_e/T\big), \quad T = 4, \qquad\qquad w_e^{(3)} = \mathbf{1}\big(g_p > 0.5\big)\, w_e. \tag{18}$$

Training therefore starts identical to the fixed graph. Section 3.3 explains why the final system keeps the fixed two-hop graph.

2.5 Symbolic Part: From IF–THEN Rules to Learned Logic

Baseline. RuleKit LR-Rules over 188 readable predicates gives about 8 crisp rules per fold. Only 10% of them mention the knowledge graph, and none of them carry a weight.

Ground atoms. All new reasoners read the same 422 facts about each patient. There are 26 clinical atoms such as $\mathit{NStage}(p, \mathrm{N3})$, 57 genomic atoms such as $\mathit{Mutated}(p, \mathrm{TP53})$, and 339 knowledge-graph atoms such as

$$\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{PI3K Cascade}).$$

The value of such an atom is its number of true groundings, the count of the patient's altered genes in pathway w:

$$n_w(p) = \big|\{\, g : \mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, w) \,\}\big|. \tag{19}$$

Atoms are chosen by prevalence (3–90% of patients), never by outcome.

1. Survival Markov logic network. Every formula has the form $w_k : \mathit{Body}_k(p) \Rightarrow \mathit{Progress}(p)$. A body is one atom or a clinical × knowledge-graph pair found on the training fold, chosen from the 25 strongest context atoms and the 60 strongest knowledge atoms by log-rank strength, up to 150 pairs. With all evidence observed and one query atom per patient, MLN inference factorises into a sum over formulas. A patient's risk is the weighted count of the formulas that hold for them:

$$\eta(p) = \sum_k w_k \, \frac{\log\big(1 + n_k(p)\big) - \mu_k}{s_k}, \tag{20}$$

where $n_k(p)$ is the number of ways formula k is true for patient p, and μ_k and s_k are its training-fold mean and standard deviation. The logarithm stops a patient with many co-amplified genes from dominating one formula.

The novelty is in how w is learned. A standard MLN would treat every patient who has not relapsed *yet* as a non-progressor. We instead maximise the Cox partial

likelihood, which uses each patient's follow-up time correctly, with an elastic-net penalty [37, 27]:

$$\max_{w} \sum_{i:\,\delta_i=1} \Big(\eta_i - \log \sum_{j:\,t_j \ge t_i} e^{\eta_j}\Big) - \lambda\Big(\alpha\|w\|_1 + \frac{1-\alpha}{2}\|w\|_2^2\Big). \tag{21}$$

We use $\alpha = 0.5$ and choose λ by an inner three-fold C-index. The ℓ_1 term sets the weight of useless formulas to exactly zero, which performs structure learning.

2. Survival PSL. The query atom is a soft value $\mathit{HighRisk}(p) = h_p \in [0, 1]$. For each of the 40 strongest facts, with counts softened to $\min(1, n/2)$, the program has a risk rule and a protective rule, plus a prior $\neg\mathit{HighRisk}(p)$ and two collective rules over each patient's ten most similar training patients:

$$\begin{aligned} w_k^+ &: \mathit{Fact}_k(p) \Rightarrow \mathit{HighRisk}(p), \\ w_k^- &: \mathit{Fact}_k(p) \Rightarrow \neg\mathit{HighRisk}(p), \\ s^+ &: \mathit{Similar}(p,q) \wedge \mathit{HighRisk}(q) \Rightarrow \mathit{HighRisk}(p), \\ s^- &: \mathit{Similar}(p,q) \wedge \neg\mathit{HighRisk}(q) \Rightarrow \neg\mathit{HighRisk}(p). \end{aligned} \tag{22}$$

Censoring is handled with soft targets. A neighbour censored at $c < 36$ months gets the Kaplan–Meier probability of progressing by 36 months, given that they were progression-free at c:

$$h_q = \frac{\hat{S}(c) - \hat{S}(36)}{\hat{S}(c)}. \tag{23}$$

Inference minimises the squared-hinge energy by 40 projected-gradient steps. Those steps are unrolled, so rule weights are learned end to end with soft-label cross-entropy plus a Cox term.

3. Reinforcement-learned rules. A GRU policy [38] writes a rule one atom at a time (up to three atoms) and is trained with REINFORCE [22] for 40,960 episodes per fold. Its reward measures *stability*. A 70% search part of the training fold is halved into A and B, and z_A, z_B are the log-rank statistics of the covered patients against the rest:

$$R = \mathbf{1}\big[\operatorname{sign} z_A = \operatorname{sign} z_B\big]\,\frac{\min(|z_A|, |z_B|, 5)}{5} - 0.03\,(\text{length} - 1). \tag{24}$$

Rules covering under 4% or over 90% of the fold earn nothing. A rule is kept only if it also separates patients on the remaining 30% of the fold, which the search never saw, with the same sign and $|z| \geq 1$.

4. RL-structured survival MLN (the engine). The rules that survive RL's filter are added to the MLN's candidate formulas, and the survival MLN learns every weight. RL does the structure learning and the survival likelihood does the weight learning.

Choosing the engine. With $k = 5$ folds, best mean C-index $\bar{C}^*$, and its fold standard deviation s^*, any reasoner above the bar below is treated as tied with the best [28]:

$$\text{bar} = \bar{C}^* - \frac{s^*}{\sqrt{k}}. \tag{25}$$

Among tied reasoners, the one with the richest reasoning becomes the engine, in the order RL-structured MLN, MLN, PSL, RL, RuleKit. The bar for the reasoners is $0.630 - 0.055/\sqrt{5} = 0.605$.

2.6 Fusion, Model Choice, and Month of Progression

The final risk comes from cross-validated stacking [39]: a Cox model with ridge penalty 0.1 on the 19 clinical fields c_i, the out-of-fold GNN risk, and the out-of-fold logic-engine risk, all standardised on the training fold:

$$\eta_i = \beta_c^\top c_i + \beta_g \, r_i^{\text{gnn}} + \beta_l \, r_i^{\text{logic}}. \tag{26}$$

It is compared with clinical-only Cox models and a random survival forest [29]. Among the models tied with the best under the same one-standard-error bar, we choose the one that can explain its prediction. As a symbolic baseline, a RuleKit patient's risk is the negative restricted mean survival time of the rules that cover them, $r_i^{\text{rule}} = -\int_0^{120} \hat{S}_i^{\text{rule}}(t)\, dt$.

Most patients never progress, so the median of their survival curve is usually undefined, and we estimate the month of progression *given* that progression happens. Two families were compared on held-out progressors. The first is the conditional median of the fused curve, the month t at which

$$\frac{F(t)}{F(\tau)} = \frac{1}{2}, \qquad F = 1 - S, \tag{27}$$

for τ of 60, 120, and 180 months. The second is an AFT model with $\mu(x) = \alpha_0 + \alpha^\top x$, fitted on training-fold progressors only with penalty 0.5, on the extended clinical design, GNN risk, and logic risk. The log-logistic form was best, so its median $e^{\mu(x)}$ is the reported month, and $t_{(0.75)}$ and $t_{(0.25)}$ give the reported range.

2.7 Explanation Paths: 0.6 Neural + 0.4 Rule

A candidate path is a real walk in the patient's fused graph: patient → sample → altered gene → (partner protein) → pathway, drug, or breast disease. Its score combines the two halves of the system:

$$\text{neural}(\pi) = \text{pct}\left(\bar{m}(\pi)\right), \tag{28}$$

$$\text{rule}(\pi) = \min\left(\frac{\left|\sum_{f \in \mathcal{F}(\pi)} w_f\right|}{S}, 1\right), \tag{29}$$

$$\text{score}(\pi) = 0.6 \times \text{neural}(\pi) + 0.4 \times \text{rule}(\pi). \tag{30}$$

Here $\bar{m}(\pi)$ is the mean GNNExplainer importance of the path's edges, and pct is its percentile among all candidate paths. $\mathcal{F}(\pi)$ holds the engine's formulas, from the fold that held the patient out, that the path grounds: the formula holds for the patient and mentions the path's end node, its altered gene, or a gene on the path. S is the 95th percentile of $|\sum w_f|$ over all rule-matched paths. The sign of $\sum w_f$ shows whether the evidence points to faster or slower progression. A path is *prominent* if its score is in the top 20% of the cohort. A report shows $K = \min(5, \text{prominent paths})$ paths and tops up to at least three.

2.8 Patient Reports

Reports are grouped by the recorded progression month: A ($\leq$12 months, 32 patients), B (12–24, 30), C (24–36, 27), D (>36, 56), and N (no progression recorded, 938). Each report is a two- to three-page PDF. It shows the clinical record, the risk percentile and the chance of progression at 12, 24, 36, and 60 months, the estimated progression month next to the recorded outcome, and the patient's survival curve. It then lists the top-K paths, the weighted formulas that fire for the patient, the molecular profile, and the most similar patients. Each group also has a summary page.

Chapter 3

Results and Analysis

3.1 Setup

All experiments ran on an 8-core laptop without a GPU, in Python with PyTorch Geometric [30], lifelines [31], scikit-survival [32], and RuleKit. Every result comes from stratified five-fold cross-validation with a fixed seed. Every prediction is *out-of-fold*: it comes from models trained on the four folds that exclude that patient. The full pipeline takes about 50 minutes, and the 1,083 reports take about 5 minutes.

3.2 Graph Neural Networks: Accuracy Saturates

Table 1 summarises the GNN benchmark on the 36-month landmark. The clinical logistic regression is the strongest model. GCN comes close, and the gene-only network is weakest. Relation-aware models and a C-index loss stayed within the fold-to-fold noise. On an external microarray cohort (GSE2034 [34]) the graph models also did not beat a linear baseline.

Table 1: 36-month progression (498 patients, 89 events), five-fold mean ± standard deviation. A random ranking has an AUPRC of 0.18.

Model	AUROC	AUPRC
Clinical logistic regression	0.703 ± 0.081	0.463 ± 0.133
GCN	0.680 ± 0.076	0.396 ± 0.091
GraphSAGE	0.650 ± 0.045	0.379 ± 0.091
GATv2	0.631 ± 0.046	0.355 ± 0.060
MLP (clinical + gene list)	0.562 ± 0.098	0.262 ± 0.064

3.3 Why Graph Structure Learning Was Not Adopted

The first GSL version closed the entire second hop: the mean edge weight was 0.24, and none of the 5,745 hop-2 edges passed 0.5. Table 2 shows that GCN's AUPRC fell as a result. The hop-prior version reported AUROC near 0.70 and AUPRC near 0.46, within the fold standard deviations. The output folder of that run was not kept, so those two numbers are quoted from its report and cannot be recomputed.

Table 2: 36-month AUPRC of the fixed two-hop graph against the first GSL version.

Model	Fixed two hops	GSL, version 1
GCN	0.396	0.316
GraphSAGE	0.379	0.383
GATv2	0.355	0.366

We left GSL out for three reasons. It gave no reliable gain. The final system's explanations rely on real, stored PrimeKG edges, and learned weights would add one more thing to justify. PyTorch Geometric's R-GCN, RGAT, GIN, and TransformerConv also accept no edge weight, so GSL could run with only three of the seven models.

3.4 Symbolic Reasoners Compared

Table 3 and Figure 3 compare the four reasoners with RuleKit on the full follow-up.

Table 3: Symbolic reasoners: five-fold C-index, formulas kept per fold, and the share of formulas that use PrimeKG.

Reasoner	C-index	Formulas/fold	Use PrimeKG
Survival MLN	0.630 ± 0.055	32.8	63%
RL-structured MLN (engine)	$\mathbf{0.628 \pm 0.055}$	35.2	**69%**
RuleKit IF–THEN rules (baseline)	0.619 ± 0.048	7.8	10%
Survival PSL	0.604 ± 0.059	40.0	63%
RL rules alone	0.563 ± 0.046	25.6	91%

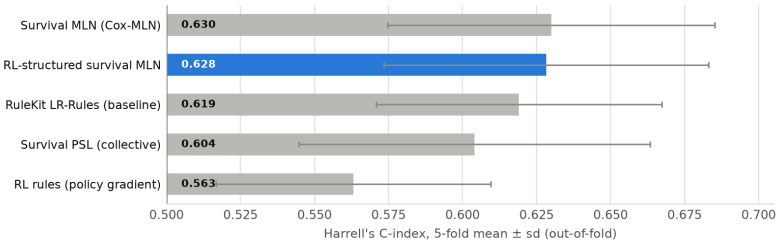


Figure 3: Mean C-index of the symbolic reasoners over five folds. The chosen engine is highlighted.

What the comparison shows.

1. **Accuracy is a tie.** Both MLN variants lead, but their gap over RuleKit (about 0.01) is smaller than the swing between folds (about 0.05). MLN, RL-structured MLN, and RuleKit are statistically tied, and the preference order selects the RL-structured MLN.

2. **The reasoning is much richer.** The engine keeps about 35 weighted formulas per fold instead of 8 crisp rules, and 69% of them reason over PrimeKG, about seven times RuleKit's share. Over the five folds it kept 176 formulas: 51 single atoms, 81 mined clinical $\times$ PrimeKG pairs, and 44 multi-step formulas written by RL.

3. **Collective reasoning found nothing.** PSL learned a weight of about 0.01 for "similar patients have similar risk". In this cohort, a patient's neighbours add no information beyond the patient's own facts.

4. **RL alone overfits.** The agent tried 20,000–27,000 different rules per fold, so

its best-looking rules are partly luck. RL works well as the formula *proposer* for the MLN, where the survival likelihood decides which proposals count.

Examples of learned formulas. Clinical facts still carry the largest weights, for example $\mathit{NStage}(p, \mathrm{N0}) \Rightarrow \mathit{Progress}(p)$ with weight -0.30 (slower progression). A mined formula raises risk ($+0.15$, 43 patients):

$$\mathit{NStage}(p, \mathrm{N3}) \wedge \mathit{Altered}(p, g) \\ \wedge \mathit{LinkedTo}(g, \text{breast ductal adenocarcinoma}) \Rightarrow \mathit{Progress}(p).$$

An RL-proposed two-pathway formula is protective (-0.14, 93 patients; Figure 4):

$$\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{VEGFA-VEGFR2}) \\ \wedge \mathit{Altered}(p, g') \wedge \mathit{InPathway}(g', \text{Ion homeostasis}) \Rightarrow \mathit{Progress}(p).$$

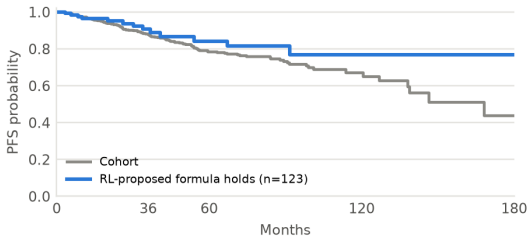


Figure 4: Kaplan–Meier curve of the patients for whom the RL-proposed VEGFA-VEGFR2 / ion-homeostasis formula holds, against the whole cohort.

A worked example. Table 4 shows the logic risk of patient TCGA-BH-A1EV (Her2, T3 N1 M0), scored by the fold that held the patient out. In the first row, $\mu = 0.051$ and $s = 0.250$ give $0.079 \times (\log 4 - 0.051)/0.250 = 0.42$. The third row shows that lacking a protective fact also raises risk relative to the average training patient: $-0.304 \times (0 - 0.331)/0.346 = +0.29$.

3.5 The Best Survival Model

Table 5 compares the main models on the full follow-up. The best mean C-index is 0.675 (clinical Cox), with one standard error of 0.016. Every model above 0.659

Table 4: Contributions to the logic risk of TCGA-BH-A1EV. Each is $w_k(\log(1 + n_k) - \mu_k)/s_k$.

Formula	w_k	n_k	Contribution
Her2 $\wedge$ altered gene in Neutrophil degranulation	+0.079	3	+0.422
Her2 $\wedge$ altered gene in L1–Ankyrins interaction	+0.055	2	+0.404
N0 stage (false for this patient)	−0.304	0	+0.290
Altered gene targeted by Ellagic acid	+0.082	1	+0.147
GOLPH3L amplified $\wedge$ altered gene in Complement cascade	+0.018	1	+0.079
All remaining formulas			+0.087
Logic risk η			**+1.429**

is statistically tied with it. Among the tied models we chose the **neuro-symbolic Cox model** (0.664) because it is the only one whose prediction can be explained with paths and formulas.

Table 5: Full follow-up (1,083 patients, 145 events): five-fold C-index and integrated Brier score to 120 months (IBS, lower is better).

Model	C-index	IBS
Clinical Cox (19 fields)	0.675 ± 0.035	0.134
Neuro-symbolic Cox (clinical + GNN + logic)	$\mathbf{0.664 \pm 0.021}$	0.134
GNN + logic, no clinical	0.642 ± 0.056	0.139
RL-structured MLN alone	0.628 ± 0.055	–
Best GNN (GraphSAGE, C-index loss)	0.622 ± 0.067	–
Random survival forest	0.554 ± 0.023	0.145

Two findings matter. First, every model family (GNN, logic, forest, and Cox) stops near a C-index of 0.67. With 145 events, stage, nodal status, age, and subtype carry most of the signal. Second, without clinical inputs the GNN and logic risks together reach 0.642, better than either alone, but once the clinical fields are present they add no independent information: in every fold their hazard ratios lie between 0.94 and 1.09 per standard deviation, with $p > 0.28$. The neuro-symbolic layer therefore adds *explanation, not accuracy.* We state this plainly in the report and in every patient report.

3.6 Month of Progression

On the 145 held-out progressors, the log-logistic AFT model is within 12 months of the recorded month for 48% of patients, against 37% for a constant guess. It places 33% of them in the correct 12-month window, against 19% for the constant guess and 25% by chance. Its median error is 12.9 months (Figure 5). Estimates are pulled towards 15–45 months: very early progressions are overestimated and very late ones underestimated. Table 6 compares the estimators. The conditional medians do poorly because proportional hazards fix the shape of the curve.

Table 6: Month estimators on 145 held-out progressors: mean and median absolute error in months, share within 12 months, and window accuracy.

Estimator	MAE	Median	$\leq$ 12 mo	Window
Constant (training progressor median)	22.9	16.0	37%	19%
Conditional median, $\tau = 60$	23.2	18.2	33%	18%
Conditional median, $\tau = 120$	27.9	24.6	23%	39%
Conditional median, $\tau = 180$	59.2	62.0	7%	39%
Log-normal AFT	20.6	12.9	48%	34%
Log-logistic AFT (chosen)	**20.1**	**12.9**	**48%**	**33%**

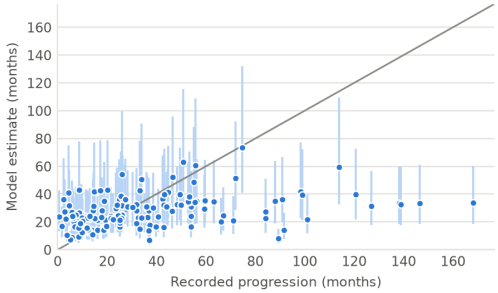


Figure 5: Estimated against recorded progression month for held-out progressors. The diagonal is perfect agreement.

3.7 Explanation Paths and Reports

The system scored 59,598 candidate paths for 1,033 patients. The other 50 patients have no altered gene with a PrimeKG link. The prominence cutoff is a score of 0.523.

35% of the paths shown in reports are backed by at least one learned formula, and a median of three formulas fires per patient. Figure 6 shows the top paths of patient TCGA-BH-A1EV (group A, progression at 12.0 months). All four are backed by formulas that raise the hazard. The strongest runs from the amplified gene NFASC to the pathway *Neurofascin interactions*. Figure 7 shows the page of the patient's report that lists these paths and the formulas that fire.

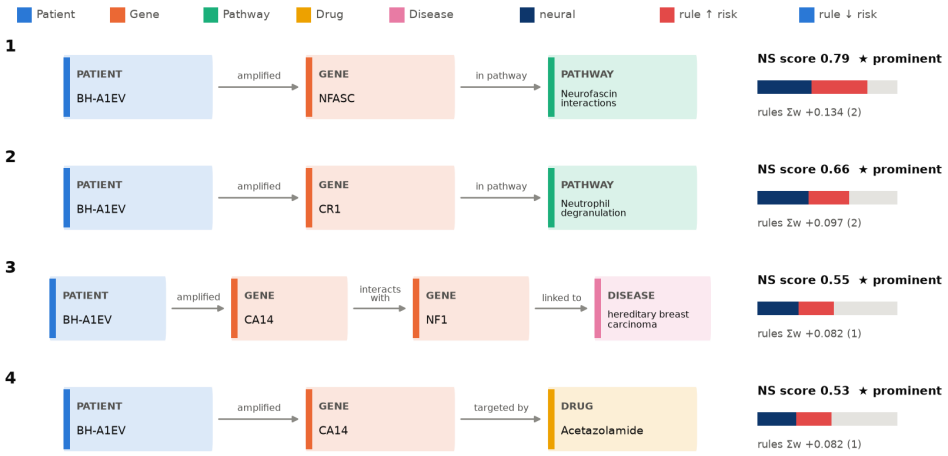


Figure 6: Top-K paths of TCGA-BH-A1EV. The bar on the right splits each score into its neural part (navy) and its rule part (red: the formulas raise the hazard).

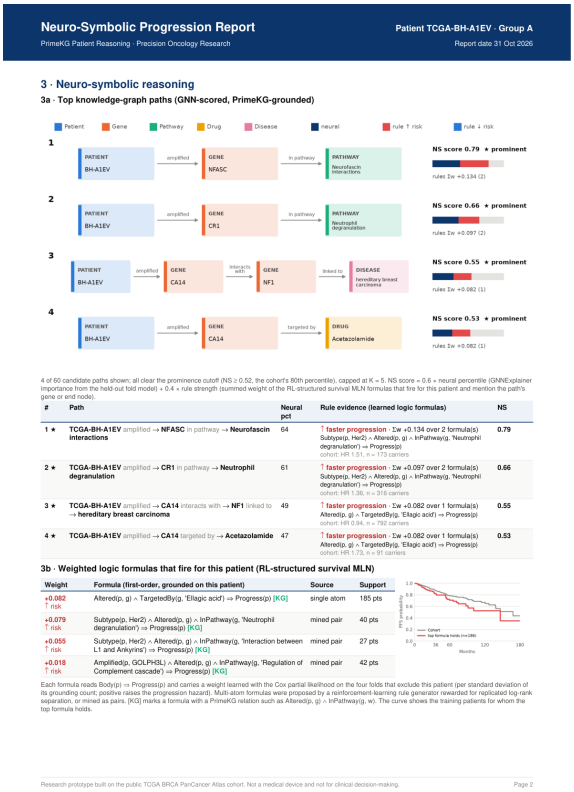


Figure 7: Page 2 of the report of TCGA-BH-A1EV: top-K knowledge-graph paths and the weighted logic formulas that fire.

Chapter 4

Conclusion and Future Work

4.1 Conclusion

We built a pipeline from raw TCGA tables to patient knowledge graphs fused with PrimeKG and benchmarked graph neural networks on them. We then turned the system into a neuro-symbolic progression model with a report for every patient. Following our supervisor's suggestion, we replaced IF–THEN rule induction with learned logic and compared Markov logic, probabilistic soft logic, and reinforcement learning. The chosen engine is a survival Markov logic network: its formulas are proposed by a reinforcement-learning agent and weighted with the Cox likelihood.

The engine matches the accuracy of the old rules and says far more. It keeps about four times as many formulas, they carry weights, and 69% of them reason over PrimeKG instead of 10%. These formulas now drive the 0.4 rule part of every explanation path. The fused model is tied with the best clinical model (C-index 0.664 against 0.675), and every prediction is out-of-fold and leakage-checked. On this cohort, knowledge-graph reasoning adds explanation rather than accuracy, and the system is designed around that finding.

4.2 Limitations

- Accuracy saturates at a C-index of about 0.67 with 145 events.

- The month estimate is coarse, and late progressions are hard to place.
- Paths show what the model relied on. They are not causal claims, and PrimeKG drug links are not treatment advice.
- The work uses one cohort and one random seed, and the reports have not been validated externally.

4.3 Future Work

- Let the RL agent walk multi-step PrimeKG relation paths and train it jointly with the MLN weights.
- Add expression and methylation data, and use pretrained knowledge-graph embeddings.
- Validate the model and the reports on an external cohort such as METABRIC.
- Ask oncologists to rate how useful and correct the reports are.

The code, the commands that reproduce every result, and two executed notebooks are in the project repository: `notebooks/neurosymbolic_patient_reports.ipynb` walks through the whole system, and `notebooks/symbolic_reasoning_novelty.ipynb` explains the logic reasoners and the novelty points.

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